

ECG Simulator Documentation

Part 5: Pathological ECG Patterns

Complete Catalog of Cardiac Arrhythmias & Conduction Abnormalities

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Contents

1 AV Block Syndrome (Group 1: Conduction Delay/Block)

1.1 First-Degree AV Block (I°)

1.1.1 Clinical Definition

Prolongation of AV conduction interval beyond upper limit of normal (200 ms), with 1:1 conduction maintained. No impulses are blocked; all atrial impulses conduct to ventricles, but with delay.

1.1.2 ECG Morphology

- **PR Interval:** ≥ 200 ms (abnormal, normal 120–200 ms)
- **P Waves:** Normal morphology and rate
- **QRS Complex:** Normal duration (≤ 120 ms)
- **Conduction Ratio:** 1:1 (every atrial beat conducts)
- **Rhythm:** Regular (no irregularity)

1.1.3 Physiological Cause

- Increased AV nodal delay (slower conduction through AV node)
- Increased refractoriness in AV nodal tissue
- Increased parasympathetic (vagal) tone
- Aging, fibrosis of AV node tissue

1.1.4 Clinical Significance

- Usually benign and asymptomatic
- May progress to higher-degree AV block in some patients
- Common in athletic individuals (vagal tone)
- Associated with: digitalis toxicity, beta-blockers, calcium channel blockers

1.1.5 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set av_delay = 0.25 s (normal 0.15 s, prolonged by 100 ms)
4. Set av_conductance = 1.0 (maintain 1:1 conduction)
5. Disable all other pathologies
6. Press Start
7. **Expected Result:** Visible PR interval ≥ 200 ms; narrow QRS; regular rhythm

1.2 Second-Degree AV Block, Mobitz Type I (Wenckebach)

1.2.1 ECG Morphology

- **PR Interval:** Progressively lengthens (200 ms $\rightarrow$ 300+ ms)
- **Dropped Beat Pattern:** P wave present but no QRS follows
- **Conduction Ratio:** Usually 5:4 or 4:3 (e.g., 5 P waves, 4 QRS complexes)
- **Rhythm:** Grouped beating; pause containing dropped beat
- **QRS Duration:** $\leq$ 120 ms (narrow QRS)

1.2.2 Physiological Cause

- Progressive fatigue of AV nodal conduction
- Decreasing AV nodal conductance over multiple cycles
- Each impulse arrives during relative refractory period
- Eventually one impulse fails to conduct (Wenckebach periodicity)

1.2.3 Clinical Significance

- Usually benign; rarely requires treatment
- Associated with: vagal tone, digitalis toxicity, ischemia
- Characteristically regular irregularity (predictable dropped beats)
- Progression to higher block is uncommon

1.2.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set av_conductance = 0.7 (partial reduction)
4. Set av_refractory = 0.35 s (slightly prolonged)
5. Gradually decrease av_conductance from 1.0 $\rightarrow$ 0.6 during observation
6. **Expected Result:** Progressive PR lengthening; periodic dropped beats; grouped rhythm

1.3 Second-Degree AV Block, Mobitz Type II

1.3.1 ECG Morphology

- **PR Interval:** Constant in conducted beats (no progressive lengthening)
- **Dropped Beats:** Sudden, without warning
- **Conduction Ratio:** Fixed (2:1, 3:1, or variable)
- **QRS Duration:** Usually $\geq$ 120 ms (wide QRS, indicating infranodal block)
- **Rhythm:** Regularly irregular (predictable dropped beats at fixed intervals)

1.3.2 Physiological Cause

- Block located in His bundle or bundle branches (infranodal)
- AV node conduction preserved; block occurs below AV node
- Abnormal automaticity or refractory period in conduction system
- Structural damage (ischemia, fibrosis, infiltration)

1.3.3 Clinical Significance

- **MORE SERIOUS than Mobitz I**
- Indicates structural disease of conduction system
- High risk of progression to complete AV block
- Often requires pacemaker implantation
- Associated with: acute MI, digitalis toxicity, cardiomyopathy

1.3.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set his_delay = 0.04 s (prolonged)
4. Set lbb_conductance = 0.5 (partial block simulates infranodal disease)
5. av_conductance = 1.0 (AV node normal; His bundle affected)
6. **Expected Result:** Fixed (not progressive) PR; sudden dropped beats; broad QRS

1.4 Third-Degree AV Block (Complete AV Block)

1.4.1 ECG Morphology

- **PR Interval:** Variable (atria and ventricles beat independently)
- **P Waves:** Normal rate and morphology (typically 60–100 bpm)
- **QRS Complexes:** Slow escape rhythm (40–60 if junctional, 20–40 if ventricular)
- **Rhythm:** Regular (atria regular; ventricles regular; unrelated to each other)
- **QRS Duration:** Depends on escape origin
 - Junctional escape: ≤ 120 ms (narrow)
 - Ventricular escape: ≥ 120 ms (broad)

1.4.2 Physiological Cause

- Complete dissociation between atrial and ventricular conduction
- Every atrial impulse blocked; no conduction through AV node or His bundle
- Ventricles must rely on backup (escape) pacemaker
- If escape fails $\rightarrow$ cardiac arrest

1.4.3 Clinical Significance

- **EMERGENT CONDITION**; requires immediate intervention
- Escape rate often inadequate for normal hemodynamics
- Symptoms: syncope, heart failure, shock
- Associated with: acute MI, digitalis toxicity, Lyme disease (tick-borne), infiltration
- Treatment: Temporary/permanent pacemaker

1.4.4 GUI Reproduction Steps

1. HR = 70 bpm (atrial rate)
2. Engine = graph (Phase 1/2)
3. Set av_conductance = 0.0 (complete block)
4. Set escape_enabled = true
5. Set escape_origin = HIS (junctional escape)
6. Set escape_interval = 1.5 s
7. **Expected Result:** P waves at 70 bpm; QRS at 40 bpm; complete dissociation

2 Bundle Branch Blocks (Group 2)

2.1 Left Bundle Branch Block (LBBB)

2.1.1 ECG Morphology

- **QRS Duration:** ≥ 120 ms (broad)
- **QRS Shape:** Notched/M-shaped in lateral leads (V5, V6, I, aVL)
- **ST Segment:** Sagging (downsloping from QRS)
- **T Waves:** Opposite to QRS polarity (typically negative in lateral leads)
- **Axis:** Usually normal to left

2.1.2 Physiological Cause

- Complete or near-complete block of left bundle branch
- Right bundle branch conducts normally
- RV depolarizes first (narrow septal Q wave)
- LV depolarizes slowly via collaterals $\rightarrow$ delayed and slow LV depolarization
- Resultant QRS vector points left and posteriorly

2.1.3 Clinical Significance

- Often indicates structural heart disease (MI, cardiomyopathy, fibrosis)
- May be associated with HF (reduced ejection fraction)
- Interferes with MI diagnosis (ST elevations may be masked)
- New LBBB = urgent evaluation required

2.1.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lbb_conductance = 0.0 (complete block)
4. Set rbb_conductance = 1.0 (normal)
5. **Expected Result:** QRS ≥ 120 ms; M-shaped QRS in V5, V6; negative T waves in lateral leads

2.2 Right Bundle Branch Block (RBBB)

2.2.1 ECG Morphology

- **QRS Duration:** ≥ 120 ms (broad)
- **QRS Shape:** M-shaped ("rsR' pattern") in V1–V2 (septal, left posterior, left anterior)
- **S Wave:** Deep and wide in I, V5, V6
- **ST Segment:** May be elevated in V1–V2 (can mimic STEMI)
- **T Waves:** Usually negative in V1–V3

2.2.2 Physiological Cause

- Complete or near-complete block of right bundle branch
- Left bundle branch conducts normally
- LV depolarizes first
- RV depolarizes slowly via collaterals $\rightarrow$ delayed RV vector rightward
- Resultant QRS vector terminates rightward (terminal R' in V1)

2.2.3 Clinical Significance

- Often normal variant (RBBB seen in 0.1% of population)
- May indicate: PE, atrial septal defect, RV strain, anterior MI
- Less ominous than LBBB if isolated
- New RBBB with symptoms warrants evaluation

2.2.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set rbb_conductance = 0.0 (complete block)
4. Set lbb_conductance = 1.0 (normal)
5. **Expected Result:** QRS ≥ 120 ms; rsR' in V1–V2; deep S in I, V5–V6

2.3 Left Anterior Fascicular Block (LAHB)

2.3.1 ECG Morphology

- **Left Axis Deviation:** Extreme left axis (-45° to -90°)
- **QRS Duration:** Typically ≤ 120 ms (narrow, unlike bundle blocks)
- **QRS Pattern:** Small R in I, Q and S in III (indicating leftward vector)
- **Clinical Finding:** Lead I shows initial R, then S wave

2.3.2 Physiological Cause

- Block of left anterior fascicle (supplies anterosuperior LV wall)
- Right and left posterior fascicles conduct normally
- Right and posterior vectors cancel; anterosuperior vector dominates
- Resultant axis points strongly leftward and superiorly

2.3.3 Clinical Significance

- Often associated with: inferior MI, degenerative disease
- May progress to complete LBB block or higher-degree AV block
- Requires baseline ECG documentation; new LAHB may indicate acute disease

2.3.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lahb_conductance = 0.0 (anterior fascicle blocked)
4. Set lphb_conductance = 1.0, rbb_conductance = 1.0 (other paths normal)
5. **Expected Result:** Extreme left axis deviation; narrow QRS; qS pattern in III

2.4 Left Posterior Fascicular Block (LPHB)

2.4.1 ECG Morphology

- **Right Axis Deviation:** Rightward axis ($+90^\circ$ to $+120^\circ$)
- **QRS Duration:** ≤ 120 ms (narrow)
- **QRS Pattern:** Small R in III, Q and S in I
- **Clinical Finding:** Rightmost QRS vector

2.4.2 Physiological Cause

- Block of left posterior fascicle (posteroinferior LV wall)
- Anterior and right fascicles conduct normally
- Inferior and posterior vectors cancel; anterior vector dominates
- Resultant axis points rightward

2.4.3 Clinical Significance

- Rare (much less common than LAHB)
- Associated with: inferior MI, infiltrative diseases
- Requires differentiation from RBBB (LPHB has narrow QRS)

2.4.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Engine = graph (Phase 1/2)
3. Set lphb_conductance = 0.0 (posterior fascicle blocked)
4. Set lahb_conductance = 1.0, rbb_conductance = 1.0
5. **Expected Result:** Right axis deviation; narrow QRS; rS pattern in I

3 Sinus Rhythm Abnormalities (Group 3)

3.1 Sinus Bradycardia

3.1.1 ECG Morphology

- **Heart Rate:** ≤ 60 bpm (normal 60–100 bpm)
- **P Waves:** Normal morphology
- **PR Interval:** Normal (120–200 ms)
- **QRS Complex:** Normal (≤ 120 ms)
- **Rhythm:** Regular (sinus rhythm maintained)

3.1.2 Physiological Cause

- Decreased SA nodal automaticity
- Increased parasympathetic tone (vagal)
- Decreased sympathetic tone
- Decreased metabolic rate (sleep, hypothermia)

3.1.3 Clinical Significance

- Often normal in athletes (resting HR 40–50 bpm)
- May indicate: hypothyroidism, increased ICP, medication effects, ischemia
- Symptomatic bradycardia: dizziness, syncope, fatigue
- Severe (<40 bpm) may require pacemaker

3.1.4 GUI Reproduction Steps

1. Set HR = 45 bpm (via HR spinner)
2. Alternatively: Set autonomic_tone = -1.0 (maximum parasympathetic)
3. **Expected Result:** Regular rhythm at 45 bpm; normal PQRST morphology

3.2 Sinus Tachycardia

3.2.1 ECG Morphology

- **Heart Rate:** >100 bpm (normal 60–100 bpm)
- **P Waves:** Normal morphology (may be small if rate very high)
- **PR Interval:** Shortened as rate increases
- **QRS Complex:** Normal (<120 ms)
- **Rhythm:** Regular

3.2.2 Physiological Cause

- Increased SA nodal automaticity
- Increased sympathetic tone
- Increased metabolic demands
- Response to exercise, fever, pain, anxiety

3.2.3 Clinical Significance

- Often physiologic (exercise, emotion, fever)
- May indicate: hyperthyroidism, anemia, heart failure, sepsis, hypovolemia
- Persistent tachycardia at rest warrants investigation
- PR shortening is expected (not abnormal)

3.2.4 GUI Reproduction Steps

1. Set HR = 150 bpm (via HR spinner)
2. Alternatively: Set autonomic_tone = +1.0 (maximum sympathetic)
3. **Expected Result:** Regular rhythm at 150 bpm; normal morphology; shortened PR

4 Atrial Arrhythmias (Group 4)

4.1 Atrial Fibrillation (AF)

4.1.1 ECG Morphology

- **P Waves:** ABSENT (replaced by baseline undulation)
- **Baseline:** Irregular oscillation (fibrillatory waves, "f waves")
- **QRS Complex:** Normal morphology (≤ 120 ms), but occurs at irregular intervals
- **Rhythm:** Completely irregular ("irregularly irregular")
- **Ventricular Rate:** Typically 100–180 bpm (fast ventricular response)
- **f-Wave Amplitude:** Variable (coarse AF 1–5 mm; fine AF ≤ 1 mm)

4.1.2 Physiological Cause

- Rapid disorganized reentry throughout atria (300–600 bpm)
- Each atrial wavelet is refractory to others $\rightarrow$ chaotic activation
- AV node bombardment by multiple disorganized impulses
- AV node acts as gatekeeper $\rightarrow$ variable conduction to ventricles
- Result: Irregular ventricular response despite chaotic atrial rate

4.1.3 Clinical Significance

- **MOST COMMON CARDIAC ARRHYTHMIA**
- Associated with: heart failure, hypertension, valvular disease, hyperthyroidism, age
- Major risk: Stroke (from atrial thrombus), hemodynamic compromise
- Treatment: Rate control (beta-blockers, CCBs, digoxin), anticoagulation, rhythm control
- Symptom severity depends on: ventricular rate, HF status, exercise tolerance

4.1.4 GUI Reproduction Steps

1. HR = 130 bpm (target controlled ventricular rate)
2. Engine = graph (Phase 1/2) or cable (Phase 3)
3. Set af_enabled = true
4. Set af_rate = 450 bpm (fibrillatory wave rate)

5. Set `af_rr_distribution = poisson` (realistic irregular RR intervals)
6. Set `af_f_amplitude = 0.05–0.10 mV`
7. **Expected Result:** Absent P waves; baseline f-waves visible; irregular QRS intervals

4.2 Premature Atrial Contraction (PAC)

4.2.1 ECG Morphology

- **P Wave:** Early (abnormal morphology, axis different from sinus)
- **PR Interval:** May be prolonged if early impulse hits refractory AV node
- **QRS Complex:** Typically normal; may be aberrant if very early
- **Coupling Interval:** 300–600 ms (time from previous QRS to premature QRS)
- **Compensatory Pause:** May be full or incomplete

4.2.2 Physiological Cause

- Premature automaticity in atrial muscle (outside SA node)
- Enhanced normal automaticity or triggered activity
- Ectopic focus in LA or RA
- Reset of SA node rhythm (typically full compensatory pause)

4.2.3 Clinical Significance

- Often benign if isolated
- Frequency may herald atrial fibrillation or other arrhythmias
- Associated with: caffeine, stress, smoking, sleep deprivation, hypoxia
- Rare: Frequent PACs may indicate atrial disease or electrolyte abnormality

4.2.4 GUI Reproduction Steps

1. HR = 70 bpm (normal sinus rate)
2. Set `ectopic_enabled = true`
3. Set `ectopic_origin = LA or RA`
4. Set `ectopic_coupling = 0.50 s` (500 ms after normal QRS)
5. Set `ectopic_repetitive = false` (isolated PAC)
6. **Expected Result:** Early P wave with different morphology; QRS follows; pause to reset rhythm

5 Ventricular Arrhythmias (Group 5)

5.1 Premature Ventricular Contraction (PVC)

5.1.1 ECG Morphology

- **QRS Complex:** Broad (>120 ms), bizarre morphology
- **QRS Axis:** Abnormal (direction depends on ectopic origin)
- **T Wave:** Opposite to QRS polarity (discordant ST-T wave)
- **ST Segment:** Depressed or elevated (opposite to QRS)
- **Coupling Interval:** 300–600 ms (time from previous beat)
- **Compensatory Pause:** Usually full (reset SA node via retrograde conduction)

5.1.2 Physiological Cause

- Premature automaticity in ventricular muscle
- Ectopic focus in LV or RV
- Ventricular depolarization begins outside normal conduction system
- Slow spread via myocardium $\rightarrow$ broad, aberrant QRS

5.1.3 Clinical Significance

- Single or rare PVCs often benign
- Frequent PVCs ($>10/\text{min}$): May indicate structural disease or electrolyte abnormality
- Patterns:
 - **Bigeminy:** Every 2nd beat is PVC
 - **Trigeminy:** Every 3rd beat is PVC
 - **Couplet:** Two consecutive PVCs
 - **Triplet/NSVT:** 3 consecutive PVCs (nonsustained VT)
- Risk factors: Hypertension, CAD, cardiomyopathy, hypoxia, hypokalemia

5.1.4 GUI Reproduction Steps

1. HR = 70 bpm
2. Set ectopic_enabled = true
3. Set ectopic_origin = LV or RV
4. Set ectopic_coupling = 0.50 s (500 ms)
5. Set ectopic_repetitive = false (single PVC)
6. **Expected Result:** Broad bizarre QRS; discordant T wave; full compensatory pause

5.2 Repetitive Ventricular Ectopy

5.2.1 Bigeminy Pattern

- **Pattern:** Normal beat — PVC — Normal beat — PVC — ...
- **Ratio:** 1:1 (equal number of normal and premature beats)
- **Rate:** If base is 70 bpm, bigeminy rate is 140 bpm
- **Clinical Significance:** Suggests more serious underlying process; warrants evaluation

5.2.2 Trigeminy Pattern

- **Pattern:** Normal — Normal — PVC — repeating
- **Ratio:** 2:1
- **Rate:** Lower than bigeminy

5.2.3 GUI Reproduction for Bigeminy

1. HR = 70 bpm
2. Set ectopic_enabled = true
3. Set ectopic_origin = LV
4. Set ectopic_coupling = 0.50 s
5. Set ectopic_repetitive = true (repetitive pattern)
6. **Expected Result:** Every 2nd beat is broad PVC; perfect 1:1 bigeminy